

1. (10 points) **Conceptual Questions.** Write an equation, an expression, or a short sentence to describe each of these concepts.

- (a) The main equation from Gauss' **Divergence Theorem** for a vector field $\vec{F}$ that relates a triple integral over a 3-dimensional bounded region (i.e., volume) T with a surface integral over its boundary surface S with an outward facing unit normal $\hat{n}$.

Solution.

$$\iint_S \vec{F} \cdot \hat{n} \, dA = \iiint_T \vec{\nabla} \cdot \vec{F} \, dV$$

□

- (b) The area element dA for the surface $\phi = \frac{\pi}{4}$ in spherical coordinates.

Solution.

$$\begin{aligned} dA &= \rho \sin \phi \, d\rho \, d\theta = \rho \sin\left(\frac{\pi}{4}\right) d\rho \, d\theta \\ &= \frac{1}{\sqrt{2}} \rho \, d\rho \, d\theta \end{aligned}$$

□

- (c) A normal vector for the sphere $x^2 + y^2 + z^2 - 4 = 0$.

Solution.

$$\begin{aligned} \vec{N} &= \vec{\nabla} (x^2 + y^2 + z^2 - 4) \\ &= 2x \hat{i} + 2y \hat{j} + 2z \hat{k} \end{aligned}$$

□

- (d) The curl of ∇f , i.e., $\nabla \times \nabla f$, where $f(r, \theta, z) = e^r \sin(\theta) - z$. *Hint:* Think before you leap!

Solution.

$$\vec{\nabla} \times \vec{\nabla} f = \vec{0} \quad (\text{Property of curl})$$

□

- (e) The expression for the volume integral $\iiint_T f(x, y, z) \, dx dy dz$ when the coordinates are changed from (x, y, z) to some other coordinates (u, v, w) .

Solution.

$$\iiint_T f(x(u, v, w), y(u, v, w), z(u, v, w)) \frac{\partial(x, y, z)}{\partial(u, v, w)} \, du dv dw$$

□

2. (10 points) Use Gauss' **Divergence Theorem** to find the surface integral $\iint_S \vec{F} \cdot \hat{n} \, dA$ for the vector field

$$\vec{F} = x\hat{i} + y\hat{j} + z\hat{k},$$

where S is the surface of a sphere of radius 2.

Solution.

Method 1: In spherical Coordinates: $\rho = \sqrt{x^2 + y^2 + z^2}$

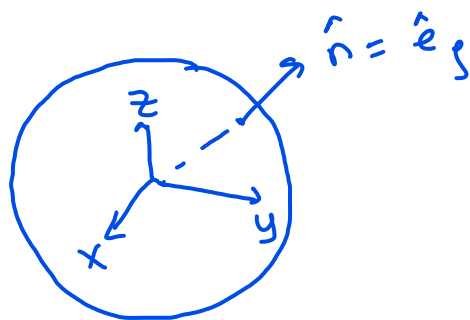
$$\vec{F} = \rho \hat{e}_\rho = F_\rho \hat{e}_\rho + \underbrace{F_\theta}_{=0} \hat{e}_\theta + \underbrace{F_\phi}_{=0} \hat{e}_\phi \quad \hat{e}_\rho = \frac{1}{\sqrt{x^2 + y^2 + z^2}} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

So $\frac{\partial}{\partial \theta} F_\theta = \frac{\partial}{\partial \phi} F_\phi = 0$

And $\vec{\nabla} \cdot \vec{F} = \frac{1}{\rho^2 \sin \phi} \frac{\partial}{\partial \rho} \left[\rho^2 \sin \phi F_\rho \right]$

$$= \frac{1}{\rho^2 \sin \phi} \frac{\partial}{\partial \rho} \left[\rho^3 \sin \phi \right]$$

$$= \frac{1}{\rho^2 \sin \phi} \cdot 3\rho^2 \sin \phi = 3$$



By Divergence Thm:

$$\iint_S \vec{F} \cdot \hat{n} \, dA = \iiint_V \vec{\nabla} \cdot \vec{F} \, dV = \int_0^{2\pi} \int_0^\pi \int_0^2 3 \underbrace{\rho^2 \sin \phi \, d\rho \, d\phi \, d\theta}_{dV}$$

$$= 3 \left(\int_0^{2\pi} d\theta \int_0^\pi \sin \phi \, d\phi \int_0^2 \rho^2 \, d\rho \right) = 3(2\pi)(2) \left(\frac{\rho^3}{3} \Big|_0^2 \right)$$

$$= 4\pi (2^3) = 32\pi$$

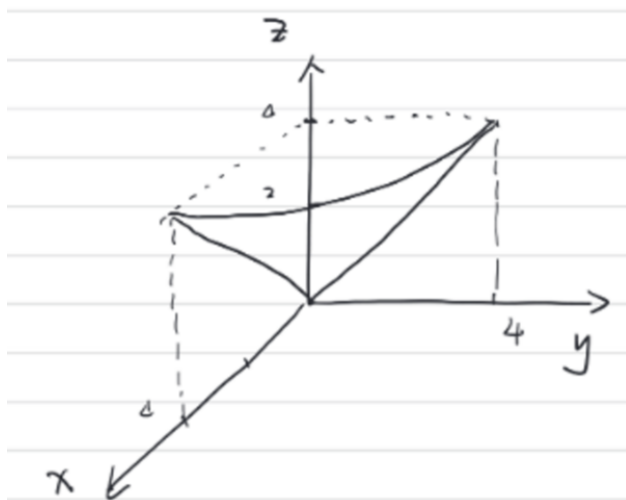
Method 2: Using Cartesian coordinates: $\vec{\nabla} \cdot \vec{F} = \frac{\partial}{\partial x} x + \frac{\partial}{\partial y} y + \frac{\partial}{\partial z} z = 3$

Divergence does not change when we change coordinates
Now integrate using spherical coordinates

$$\iint_S \vec{F} \cdot \hat{n} \, dA = \iiint_V \vec{\nabla} \cdot \vec{F} \, dV = \iiint_V 3 \, dV = 32\pi \text{ as in method 1.}$$

□

3. (10 points) Evaluate $\iiint_T (x + y + z) \, dx \, dy \, dz$ where T is the volume of the first octant inside the cone $x^2 + y^2 \leq 4z^2$, $z \in [0, 2]$, as shown below.

Figure 1: Region T .

Solution.

Using Cylindrical Coordinates: $T: r^2 \leq 4z^2, z \in [0, 2]$

$$\begin{aligned}
 &\text{Then} \\
 &\iiint_T (x + y + z) \, dx \, dy \, dz = \int_0^2 \int_0^{4z^2} \int_0^{2\pi} (r \cos \theta + r \sin \theta + z) \, d\theta \, dr \, dz \\
 &= \int_0^2 \int_0^{4z^2} \left[r \sin \theta - r \cos \theta + z\theta \right]_0^{2\pi} \, dr \, dz \\
 &= \int_0^2 \int_0^{4z^2} 2\pi z \, dr \, dz = \int_0^2 \left[2\pi z r \right]_0^{4z^2} \, dz \\
 &= \int_0^2 2\pi z (4z^2) \, dz = \int_0^2 8\pi z^3 \, dz = \left[\frac{8\pi z^4}{4} \right]_0^2 \\
 &= 2\pi (2)^4 = 32\pi
 \end{aligned}$$

□

4. (10 points) Verify Stokes' Theorem by computing the following line integral:

$$\oint_C \vec{F} \cdot d\vec{r},$$

where $\vec{F} = -5y\hat{i} + 4x\hat{j} + z\hat{k}$, and C is the circle $x^2 + y^2 = 16$, $z = 4$, traversed counter-clockwise, and then comparing this result to that of the surface integral

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, dA,$$

where S is the surface $x^2 + y^2 \leq 4$, $z = 4$.

Solution.

Parameterize: $C: \vec{r} = [4\cos t, 4\sin t, 4]$
 $\Rightarrow \vec{r}' = [-4\sin t, 4\cos t, 0]$

$$\vec{F} = -5y\hat{i} + 4x\hat{j} + z\hat{k} = -5(4\sin t)\hat{i} + 4(4\cos t)\hat{j} + 4\hat{k}$$

$$\Rightarrow \vec{F} = -20\sin t\hat{i} + 16\cos t\hat{j} + 4\hat{k}$$

$$\text{Then } \vec{F} \cdot \vec{r}' = 80\sin^2 t + 64\cos^2 t + 0$$

$$= 16\sin^2 t + 64\sin^2 t + 64\cos^2 t = 16\sin^2 t + 64$$

$$\begin{aligned} \text{So } \oint_C \vec{F} \cdot d\vec{r} &= \int_0^{2\pi} (16\sin^2 t + 64) dt = \int_0^{2\pi} 8(1 - \cos 2t) + 64 \, dt \\ &= \int_0^{2\pi} 72 - \cos 2t \, dt = \left[72t - \frac{1}{2}\sin 2t \right]_0^{2\pi} \end{aligned}$$

$$= 72(2\pi) - 0 = 144\pi$$

* Surface integral on page 5: .

□

5. (10 points) In the Cartesian plane with coordinates (x, y) , we may define another set of coordinates (u, v) using the transformation functions

$$x = u^2 + v^2, \quad y = u^2 - v^2.$$

Find both the Lamé coefficients h_u and h_v and thus provide the expression for the line element ds^2 and the area element dA in this new coordinate system.

Solution. $x = u^2 + v^2$ $y = u^2 - v^2$

$$\frac{\partial x}{\partial u} = 2u \quad \frac{\partial x}{\partial v} = 2v$$

$$\frac{\partial y}{\partial u} = 2u \quad \frac{\partial y}{\partial v} = -2v$$

$$\text{Then } h_1^2 = h_u^2 = \left(\frac{\partial x}{\partial u}\right)^2 + \left(\frac{\partial y}{\partial u}\right)^2 = (2u)^2 + (2u)^2 = 8u^2$$

$$h_2^2 = h_v^2 = \left(\frac{\partial x}{\partial v}\right)^2 + \left(\frac{\partial y}{\partial v}\right)^2 = (2v)^2 + (-2v)^2 = 8v^2$$

$$\Rightarrow h_u = \sqrt{8} u, \quad h_v = \sqrt{8} v$$

$$\text{Then } ds^2 = h_u^2 du^2 + h_v^2 dv^2 = 8u^2 du^2 + 8v^2 dv^2$$

$$dA = h_u h_v du dv = (\sqrt{8} u)(\sqrt{8} v) du dv = 8uv du dv$$

* Surface integral from problem 4:

Note that the surface $S: x^2 + y^2 = 16, z = 4$ is parallel to xy -plane so $\hat{k}$ is a normal vector. So for $(\vec{\nabla} \times \vec{F}) \cdot \hat{n}$, we only need the $\hat{k}$ component of $\vec{\nabla} \times \vec{F} \Rightarrow \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = \frac{\partial(4x)}{\partial x} - \frac{\partial(-5y)}{\partial y} = 9$

$$\text{Thus } \iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} dA = \iint_S 9 dA = 9(\text{area of circle of radius 4})$$

$$= 9(\pi(4)^2) = 9(16)\pi = 144\pi$$

Same answer as $\oint_C \vec{F} \cdot d\vec{r} \Rightarrow \text{Stokes' verified. } \square$

Formula Sheet

General Coordinates:

$$(x_1, x_2, x_3) \rightarrow (q_1, q_2, q_3):$$

$$x_1 = x_1(q_1, q_2, q_3), \quad x_2 = x_2(q_1, q_2, q_3), \quad x_3 = x_3(q_1, q_2, q_3)$$

$$\text{Derivative Matrix: } D = \begin{bmatrix} \frac{\partial x_1}{\partial q_1} & \frac{\partial x_1}{\partial q_2} & \frac{\partial x_1}{\partial q_3} \\ \frac{\partial x_2}{\partial q_1} & \frac{\partial x_2}{\partial q_2} & \frac{\partial x_2}{\partial q_3} \\ \frac{\partial x_3}{\partial q_1} & \frac{\partial x_3}{\partial q_2} & \frac{\partial x_3}{\partial q_3} \end{bmatrix}$$

$$\text{Lamé coefficients: } h_j^2 = \sum_{k=1}^3 \left(\frac{\partial x_k}{\partial q_j} \right)^2$$

$$\text{Jacobian: } \frac{\partial(x_1, x_2, x_3)}{\partial(q_1, q_2, q_3)} = |J| = |\det D| = |h_1 h_2 h_3|$$

$$\text{Line element: } ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2 + h_3^2 dq_3^2$$

$$\text{Volume element: } dV = h_1 h_2 h_3 dq_1 dq_2 dq_3$$

$$\text{Area element for surface with } q_j = \text{const } (i, j, k \text{ distinct}): dA = h_i h_k dq_i dq_k$$

$$\text{Gradient for scalar function } f: \nabla f = \frac{1}{h_1} \frac{\partial f}{\partial q_1} \hat{e}_{q_1} + \frac{1}{h_2} \frac{\partial f}{\partial q_2} \hat{e}_{q_2} + \frac{1}{h_3} \frac{\partial f}{\partial q_3} \hat{e}_{q_3}$$

$$\text{Divergence and curl of vector field } \vec{F} = F_{q_1} \hat{e}_{q_1} + F_{q_2} \hat{e}_{q_2} + F_{q_3} \hat{e}_{q_3}:$$

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} (h_2 h_3 F_{q_1}) + \frac{\partial}{\partial q_2} (h_3 h_1 F_{q_2}) + \frac{\partial}{\partial q_3} (h_1 h_2 F_{q_3}) \right]$$

$$\nabla \times \vec{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{e}_{q_1} & h_2 \hat{e}_{q_2} & h_3 \hat{e}_{q_3} \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ h_1 F_{q_1} & h_2 F_{q_2} & h_3 F_{q_3} \end{vmatrix}$$

Laplacian of scalar field f :

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right].$$

Cylindrical Coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

Spherical Coordinates:

$$x = \rho \cos \theta \sin \phi, \quad y = \rho \sin \theta \sin \phi, \quad z = \rho \cos \phi$$

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$